Unit 2 Using the System

Course code LX13
Linux Basics

Objectives

After completing this unit, students should be able to:

- Log in and out of the system
- State the structure of Linux commands
- Execute basic Linux commands
- Use Linux commands to communicate with other users

Logging In and Out

- To log in
  host1 login: **tux1**
  Password: (the password does not appear)
  Last Login: Fri Feb 26 17:35:14 on tty1
  [tux1@host1 tux1]$

- To log out
  `$ <ctrl-d>`
  or
  `$ exit`
  or
  `$ logout`

Passwords

- Creating or Changing

```
$ passwd
Changing password for tux1
(current) UNIX password:
New UNIX password:
Retype new UNIX password:

$
```

Command Format

- Linux commands have the following format:
  $ command option(s) argument(s)

- For example:

```
$ ls
$ ls -l
$ ls /dev
$ ls -l /dev
```

Command Format Examples

RIGHT	WRONG
1. Separation	
`$ mail -f personal`	`$ mail - f personal`
`$ who -u`	`$ who-u`
2. Order	
`$ mail -f personal`	`$ mail personal -f`
`$ who -u`	`$ -u who`
3. Multiple options	
`$ who -m -u`	`$ who -m-u`
`$ who -mu`	`$ who -m u`

The date and cal Commands

- Checking the date

```
$ date
Fri Mar  5  11:15:10 CET 1999
```

- Looking at a month

```
$ cal 3 1999
        March 1999
Su  Mo  Tu  We  Th  Fr  Sa
     1   2   3   4   5   6
 7   8   9  10  11  12  13
14  15  16  17  18  19  20
21  22  23  24  25  26  27
28  29  30  31
```

- Looking at a year

```
$ cal 1999
```

Who Is on the System

- Finding who is on the system

```
$ who
root   tty1     Mar  5  11:10
peter  tty2     Mar  5  11:04

$ finger
Login  Name  Tty Idle  Login   Time
peter          2       Mar  5  11:04
root   root    *1   7   Mar  5  11:10
```

Finding Information about Users

- Finding who you are

```
$ who am i
host!peter tty2    Mar  5  11:04
```

or

```
$ whoami
peter
```

- The finger command

```
$ finger peter
Login: peter            Name:
Directory: /home/peter  Shell: /bin/bash
On since Fri Mar  5  11:04 (CET) on tty2
No mail.
No plan.
```

The clear and echo Commands

- The clear command
  The clear command clears the terminal screen

```
$ clear
```

- The echo command
  The echo command writes the arguments to the screen

```
$ echo Hopefully, lunch is at 12:00
Hopefully, lunch is at 12:00
```

The wc Command

- The wc command counts the number of lines, words and bytes in a named file.

```
$ wc [-l][-w][-c] filename
```

-l counts the number of lines
-w counts the number of words
-c counts the number of bytes (characters)

```
$ wc  .bash_profile
15   31   230   .bash_profile
```

Sending Mail

$ mail allet

Subject: Meeting

There will be a brief announcement meeting today.

<ctrl-d>

Cc: <enter>

$ mail team02@host1

Subject: Don't forget!

Don't forget this Linux course!!!

<ctrl-d>

Cc: <enter>

Receiving Mail

```
You have mail in /var/spool/mail/allet
$ mail
Mail version 8.1 6/6/93  Type ? for help
"/var/spool/mail/allet": 2 messages 1 new 2 unread
 U  1 peter@host  Thu Mar  4  16:10 35/1185 "Status"
>N  2 peter@host  Fri Mar  5  11:34 13/350 "Meeting"
& 2
Message 2:
From peter  Fri Mar  5  11:34:10 199
Date: Fri, 5 Mar 1999  11:34:09 +0100
To: allet@host
Subject: Meeting

There will be a brief announcement meeting today.
& q
```

Write and Wall

- Use write to display a text message on a user's terminal

```
$ write team02
Message
<ctrl-d>
```

- Use wall to place a message on all logged in user's displays

```
$ wall
I'm back
<ctrl-d>
```

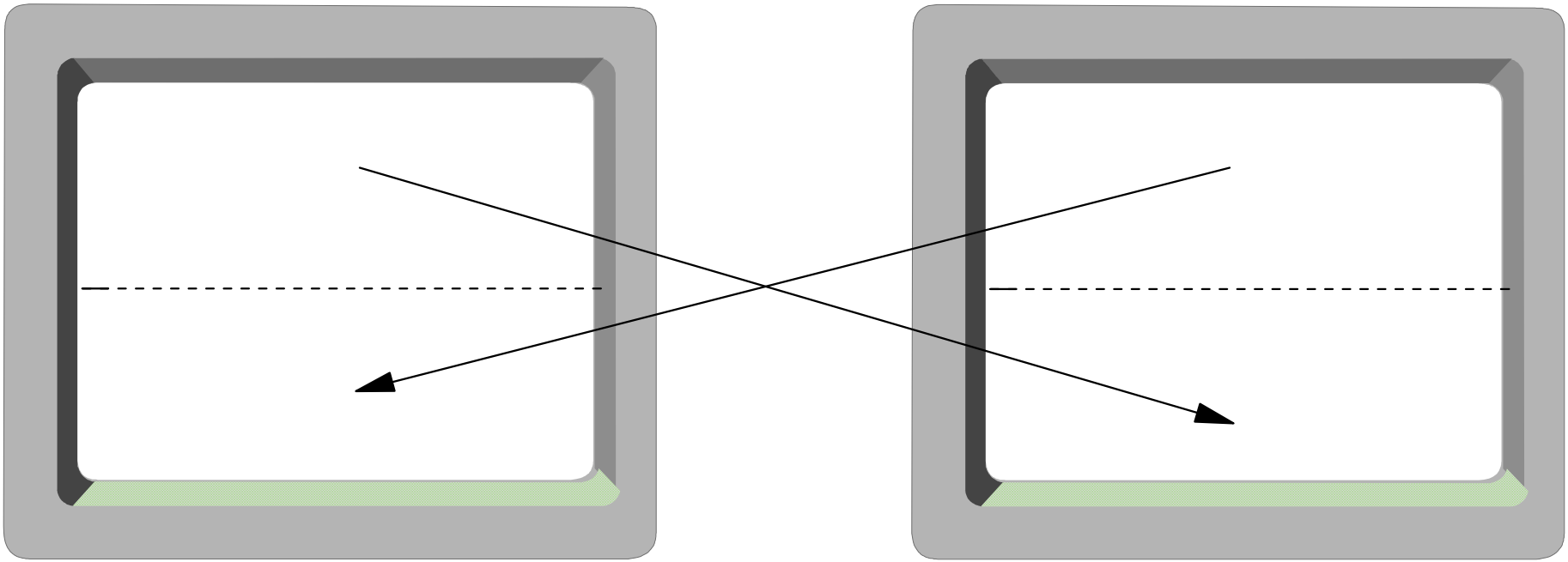
Talk with Another User

If John wants to talk with Fred, John enters:

```
$ talk fred
```

If Fred also wants to talk with John, Fred enters:

```
$ talk john
```



The mesg Command

- The mesg command controls whether other users can send messages to you with the **write, wall** or **talk** command or through **output redirection**.

```
$ mesg [y|n]
```

- The root user can override the default permission settings.

Keyboard Tips

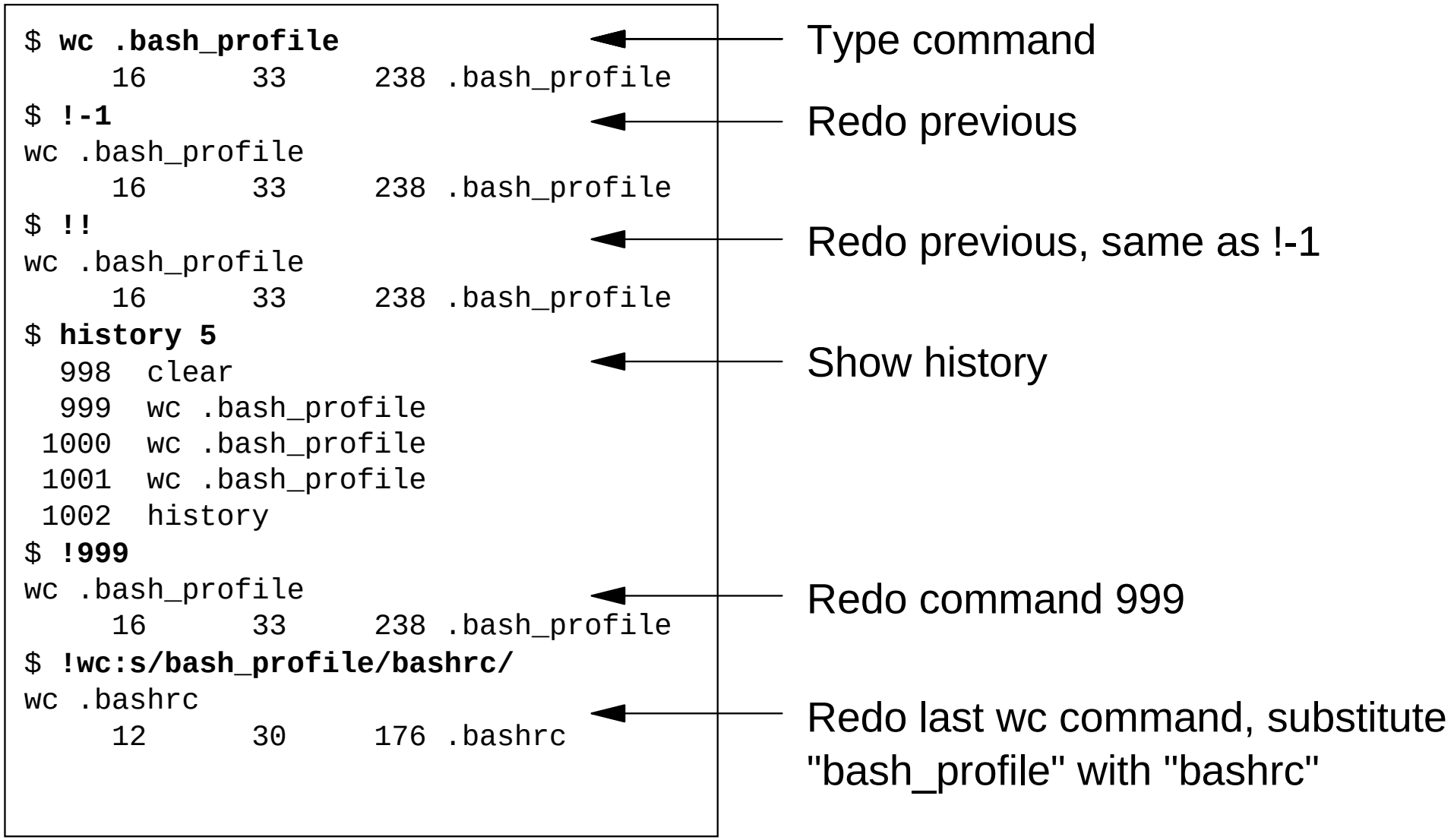
- <backspace>
- Corrects mistakes
- <ctrl-c>
- Terminates the current command and returns to the shell
- <ctrl-d>
- End of transmission or end of line
- <ctrl-s>
- Temporarily stops output to the screen
- <ctrl-q>
- Resumes output
- <ctrl-u>
- Erase the entire line

Keyboard Tips *(2)*

- <arrow up>
- Previous command
- <arrow down>
- <arrow left>
- <arrow right>
- <shift page-up>
- Scroll page up
- One character to the left
- <shift page-down>
- One character to the right
- Look at the output of previous commands
- Go back to your prompt

Command History

- Command history can be viewed with **history** command
- To retrieve commands, use **!**



Checkpoint

1. What is the correct command syntax in Linux?
   a. $ mail newmail -f
   b. $ mail f newmail
   c. $ newmail -f mail
   d. $ mail -f newmail

2. What commands can be used to communicate with other users?

_____

Checkpoint *(2)*

3. Which commands would you use to find out when a particular user logged in?
   a. $ who am i
   b. $ whoami
   c. $ who
   d. $ finger

_____

4. What key sequence would you use to recall a command from history?

_____

Unit Summary

- Linux commands can use multiple options and arguments and must follow proper syntax rules.

- There are many simple, yet powerful commands such as:
  - date
  - cal
  - who, who am i, whoami
  - finger
  - echo
  - clear
  - wc

- Communicate with other Linux users using commands such as mail, write, wall and talk.